10" Homelab Rack Raspberry Pi 4 Modular Rack

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VIEW IN BROWSER

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Summary

Modular Raspberry Pi holder for 10" Homelab Rack

Gadgets > Computers

Tags: raspberry rack ssd pi4 homelab

Designed to suit my own needs alongside other models I've uploaded on Printables

Raspberry Pi4 holder with or without a SSD, one or two devices per 1U rack space. I used the GeeekPi DeskPi KL-P24 Raspberry Pi Adaptor Board to get a front facing USB-C and Full Size HDMI ports.

There is enough room above the Pi' for a HAT (1 of mine is using a PoE+ HAT) but I've left my heatsinks on my Pi's

4 Different Prints:

1. The 1U Blade to hold the Raspberry Pi
2. Raspberry Pi4 holder (with slot for SSD)
3. A blanking plate if you're only installing one Pi
4. A blanking plate for SSD if you're using SD card

You'll need threaded inserts to take full advantage of the build:

- The DeskPi KL-P24 uses 2x M2.5 Inserts
- The 1U Blade uses 4x M3 Inserts

You'll also need the M3 bolts to fasten the devices into the 1U Blade (you get the M2.5 screws with the DeskPi board)

Design issues (being open):

I used to have all my Raspberry Pi's sat on my desk using GeeekPi Alloy cases, the first one I designed the Pi4 Holder around had smaller screws than the other 2 so the holes needed some work for me.

The SSD blank is quite tight so the hex flexes a little when installed not an issue to the purpose and can be removed without issue and reused if needed.

I've printed all mine in PETG (Galaxy Black) with 15% infil, no brim on the 1U Blade and organic supports on models that need it.

Model files

rpi4-plate-v6.stl

rpi4-plate-v6.3mf

rpi4-ssd-v2-v3.3mf

rpi4-ssd-v2-v3.stl

rpi4-blankpanel-v4.stl

rpi4-blankpanel-v4.3mf

rpi4-ssd-blank-v4.3mf

rpi4-ssd-blank-v4.stl

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